# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 8: **Heredity** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The number of chromosomes in a human sperm is 23 and in the egg is 23. In a normal human zygote and a skin cell, the numbers are respectively 1

**(a)** 46 and 46 **(b)** 23 and 46 **(c)** 23 and 23 **(d)** 46 and 23

**2.** Assertion (A): Variations are important for the survival of species. Reason (R): They provide a chance to adapt to changing conditions. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**3.** The probability that a couple's four children are all girls is 1

**(a)** 1/2 **(b)** 1/4 **(c)** 1/8 **(d)** 1/16

**4.** A population of asexually reproducing bacteria shows less variation than a population of humans because 1

**(a)** bacteria have no DNA
**(b)** sexual reproduction reduces variation
**(c)** humans have no mutation
**(d)** there is only one parent and only copying errors

**5.** Assertion (A): Children resemble their parents but are not identical. Reason (R): They inherit genes from both parents in new combinations. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**6.** Of the children of a couple, which parent's chromosome decides that a son carries a Y? 1

**(a)** Mother **(b)** Father **(c)** Both **(d)** Neither

**7.** A plant with TtRr genotype is self-pollinated. The probability of an offspring being dwarf and wrinkled is 1

**(a)** 1/4 **(b)** 1/16 **(c)** 3/16 **(d)** 9/16

**8.** Assertion (A): A test cross is carried out with a recessive parent. Reason (R): Recessive parent produces only one type of gamete. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**9.** **Assertion (A):** A recessive trait appears in a plant only when it has two copies of the recessive allele. 1
**Reason (R):** A dominant allele masks the effect of a recessive allele in the heterozygote.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**10.** **Assertion (A):** A cross between a round-seeded plant (Rr) and a wrinkled-seeded plant (rr) gives round and wrinkled seeds in the ratio 1 : 1. 1
**Reason (R):** When pure round and pure wrinkled pea plants are crossed, all F1 seeds are round.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** If both parents are TT and tt, what is the probability that the F2 is tt? Explain. 2

**12.** Find the probability that a child is a girl if the father's sperm carries a Y chromosome. 2

**13.** How many types of gametes will be produced by TtRrYy? Show the reasoning. 2

**14.** Why can't an individual with recessive phenotype be heterozygous? 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Explain the fallacy in the statement: A man who exercises regularly will have sons with strong muscles. 3

**16.** A tall plant (Tt) is self-pollinated and gives 4 offspring. Can we say one will be dwarf? Explain. 3

**17.** What is the effect of the environment on a person's height even if the genes for height are the same? 3

OR

Why is the chance of the first child being a boy the same as the chance of the second child being a boy? 3

**18.** Study the statement: Mendel's F1 had no dwarf plants, so the dwarf trait was lost. Evaluate. 3

OR

A sperm contains 22 autosomes and X. Name the sex of the child and explain how the child gets its other sex chromosome. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** A single bacterium divides to form two bacteria, and each of these divides again. The four bacteria are very similar but show slight differences due to small errors in DNA copying. In humans, who reproduce sexually, much greater variations are seen among individuals. During a heat wave, only some of the bacteria with a particular variation survive. 5
(i) What is meant by heredity?
(ii) What causes variation in asexual reproduction?
(iii) Why is variation greater in sexual reproduction?
(iv) Explain why variation is useful for the survival of a species.

**20.** Explain how Mendel's experiments and the idea of genes on chromosomes together explain why the F2 of a dihybrid cross has a 9 : 3 : 3 : 1 ratio. 5

OR

In humans, Rh positive is dominant over Rh negative. A heterozygous woman marries a Rh negative man. Show the possible children with their genotypes, phenotypes and the probability of each. 5

## END OF QUESTION PAPER — Total: 40 Marks